InsightAI Analytics Report

SuperMarket Analysis.csv

Generated on March 10, 2026 09:39 PM

Rows	**1,000**
Columns	**17**
Missing values handled	**0**
Duplicates removed	**0**
Anomalies detected	**0**

This dataset contains 1,000 rows across 17 columns. Top performing Product line is Food and beverages with total Sales of 56,144.84. Sales has the highest average value at 322.97 across the numeric columns. Sales shows a decreasing trend across month-level Date data, moving from 116,291.87 to 109,455.51 and peaking at 116,291.87 in 2019-01. Strongest correlation is positive between Tax 5% and gross income (1.00).

Prepared for data exploration, stakeholder reporting, and decision-ready business review.

1. Dataset Summary

This section captures the structure of the active dataset after validation, missing-value handling, and duplicate removal.

Dataset name	SuperMarket Analysis.csv
Rows	1,000
Columns	17
Numeric columns	Unit price, Quantity, Tax 5%, Sales, cogs, gross margin percentage, gross income, Rating
Categorical columns	Invoice ID, Branch, City, Customer type, Gender, Product line, Payment
Datetime columns	Date, Time

No anomalies were detected at the configured Z-score threshold.

2. Statistical Analysis

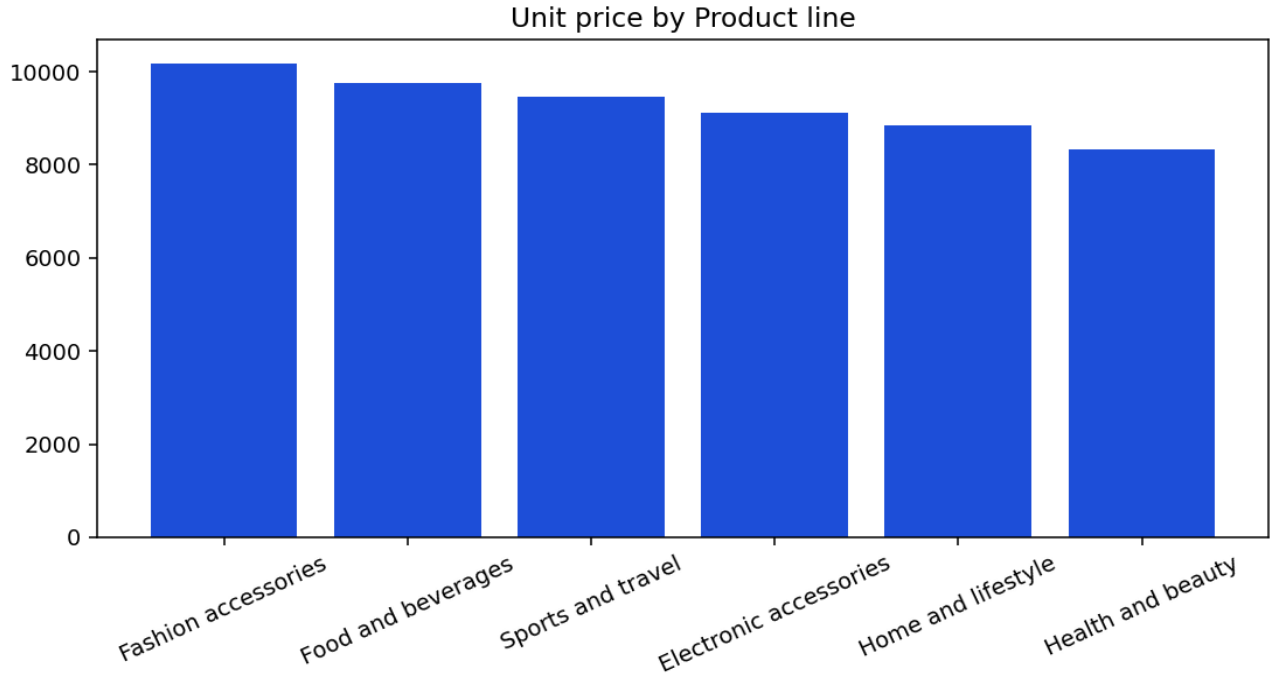
Descriptive statistics below highlight central tendency, spread, and range across the leading numeric fields.

Column	Mean	Median	Mode	Min	Max	Std Dev
Unit price	55.6721	55.23	83.77	10.08	99.96	26.4814
Quantity	5.51	5.0	10.0	1.0	10.0	2.922
Tax 5%	15.3794	12.088	4.154	0.5085	49.65	11.703
Sales	322.9667	253.848	87.234	10.6785	1042.65	245.7624
cogs	307.5874	241.76	83.08	10.17	993.0	234.0594
gross margin percentage	4.7619	4.7619	4.7619	4.7619	4.7619	0.0
gross income	15.3794	12.088	4.154	0.5085	49.65	11.703
Rating	6.9727	7.0	6.0	4.0	10.0	1.7177

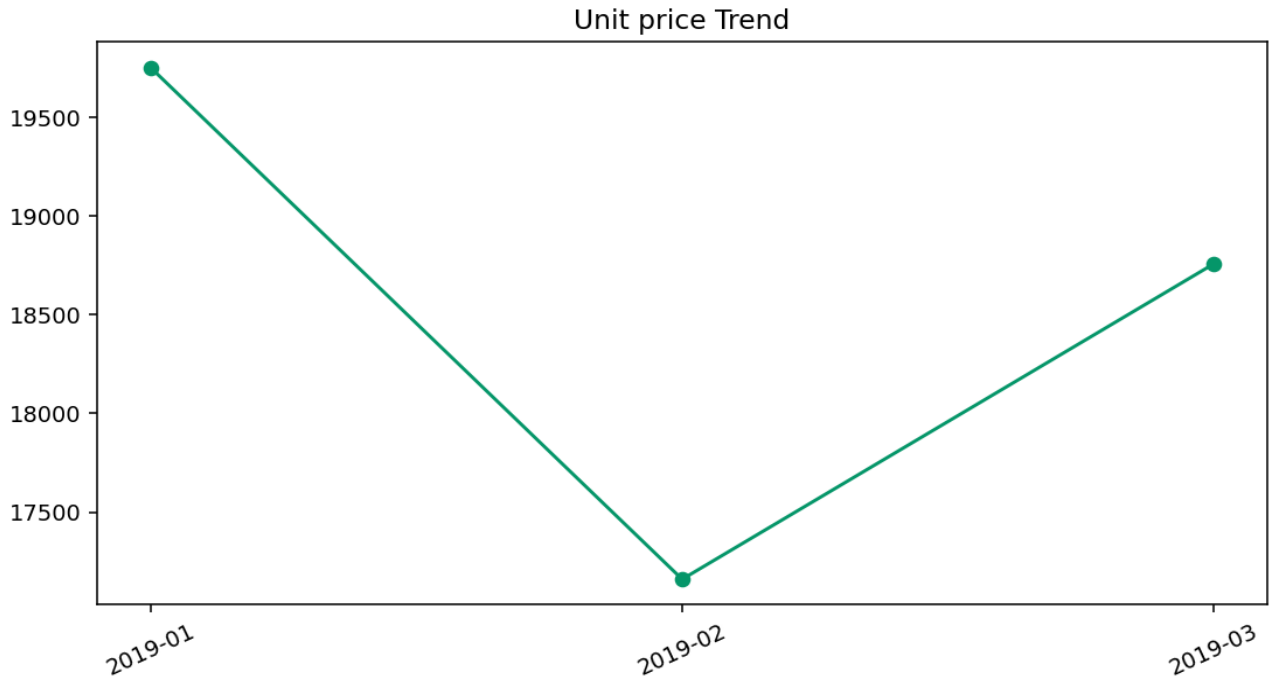
3. Charts

The following visualizations are generated automatically from the uploaded dataset using matplotlib and embedded directly in the report.

Category Performance

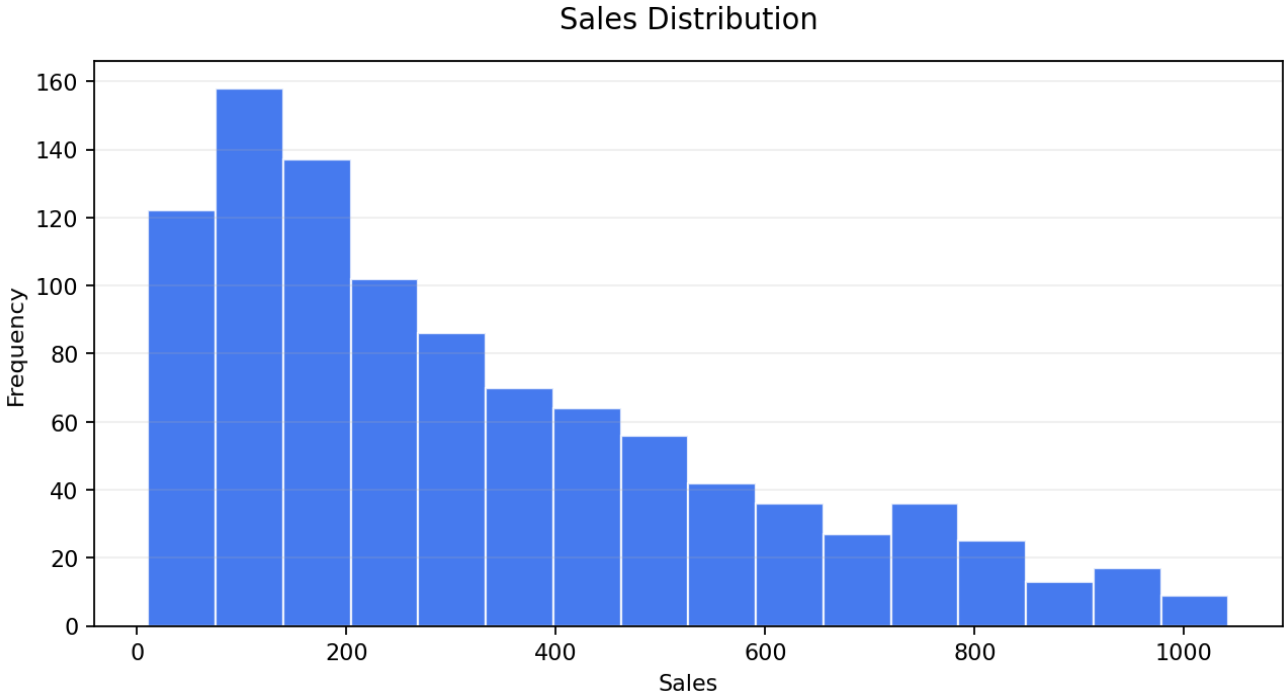


Time Trend

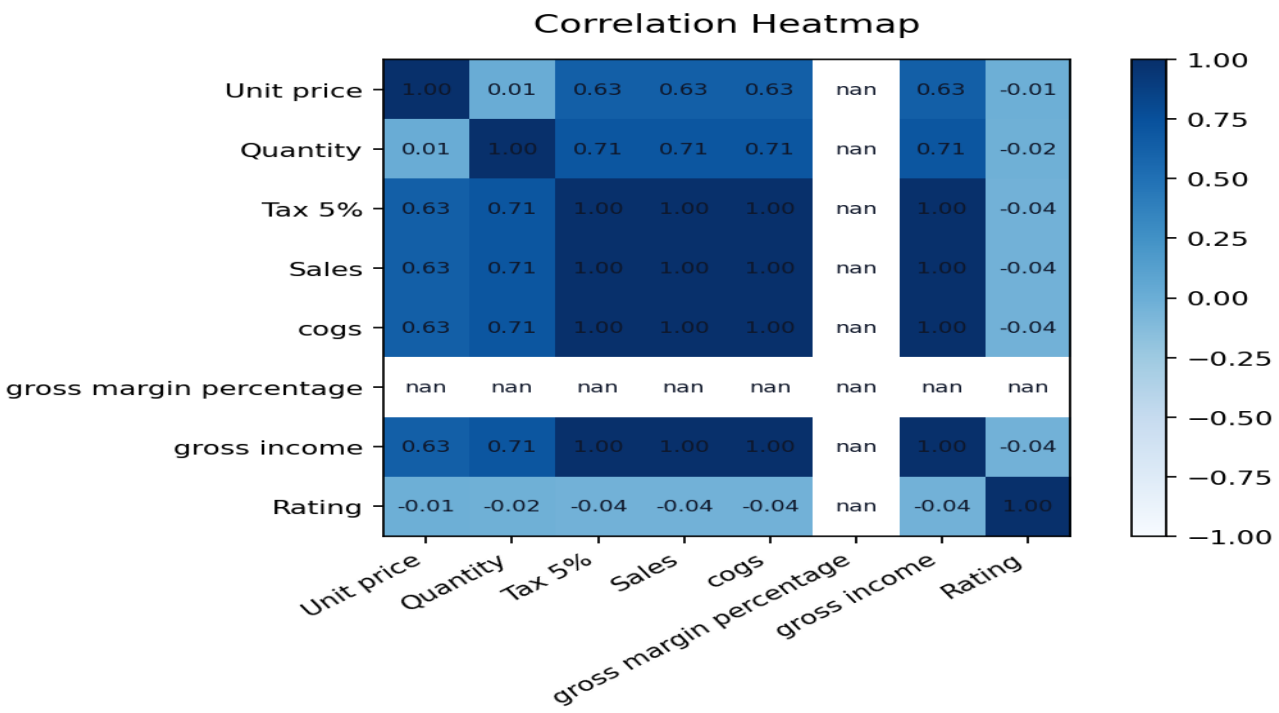


Scatter Analysis

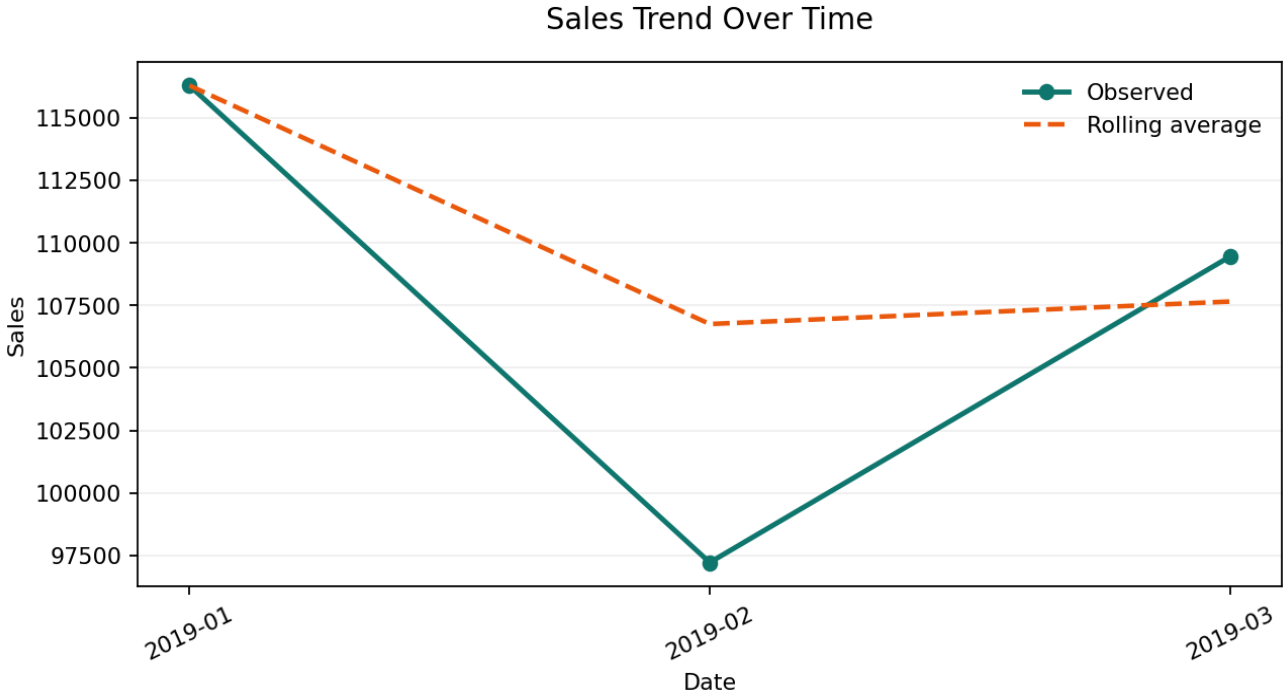
Sales Distribution



Correlation Heatmap



Sales Rolling Trend



4. AI Insights

This dataset contains 1,000 rows across 17 columns. Top performing Product line is Food and beverages with total Sales of 56,144.84. Sales has the highest average value at 322.97 across the numeric columns. Sales shows a decreasing trend across month-level Date data, moving from 116,291.87 to 109,455.51 and peaking at 116,291.87 in 2019-01. Strongest correlation is positive between Tax 5% and gross income (1.00).

Highlights

- Top performing Product line is Food and beverages with total Sales of 56,144.84.
- Sales has the highest average value at 322.97 across the numeric columns.
- Columns with the highest variance are Sales (60,459.60), cogs (54,838.64), and Unit price (701.97), indicating where the data fluctuates the most.

Trends

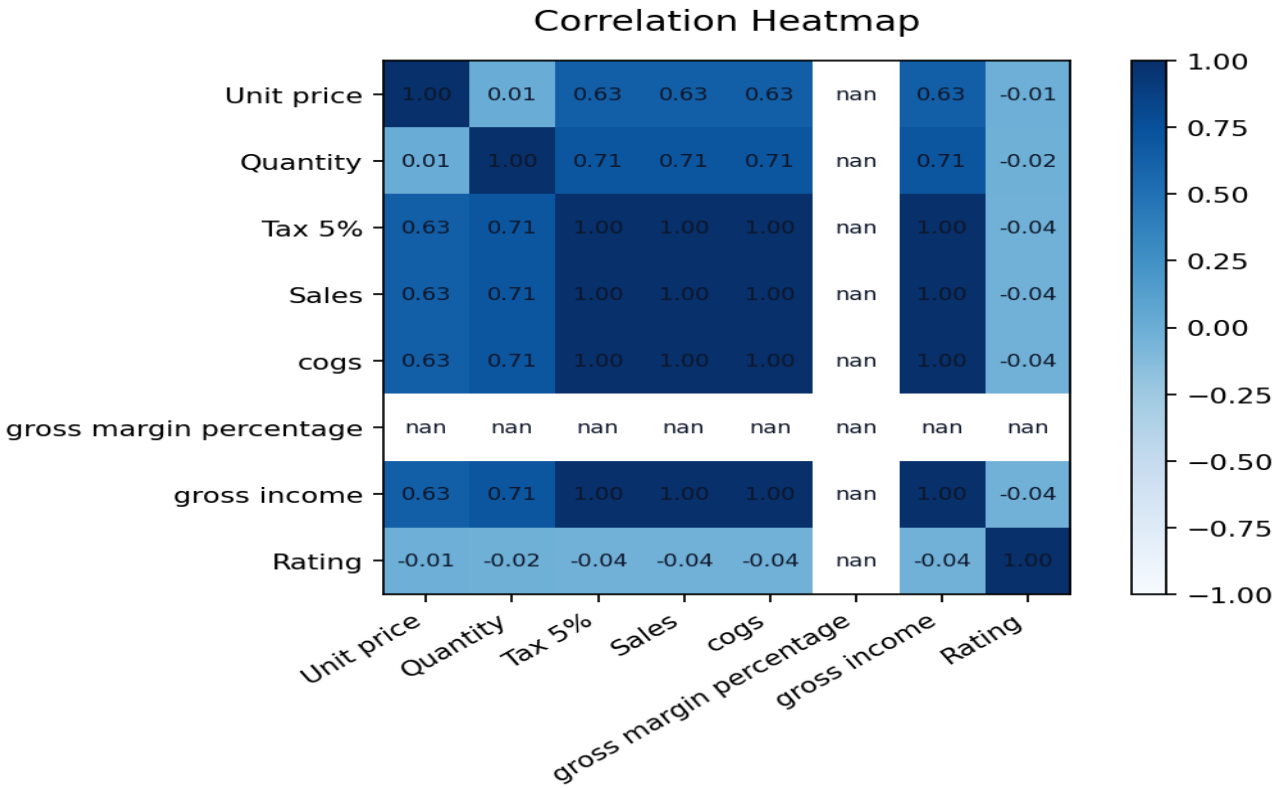
- Sales shows a decreasing trend across month-level Date data, moving from 116,291.87 to 109,455.51 and peaking at 116,291.87 in 2019-01.
- Unit price remains relatively stable across month-level Date data, with a peak mean of 56.63 in 2019-02.

Correlations

- Strongest correlation is positive between Tax 5% and gross income (1.00).
- Tax 5% and cogs also show a strong positive relationship (1.00).
- cogs and gross income also show a strong positive relationship (1.00).

5. Correlation Analysis

Strongest correlation is positive between Tax 5% and gross income (1.00).

Variable 1	Variable 2	Correlation
Tax 5%	gross income	1.00
Tax 5%	cogs	1.00
cogs	gross income	1.00
Tax 5%	Sales	1.00
Sales	gross income	1.00
Sales	cogs	1.00
Quantity	Tax 5%	0.71
Quantity	gross income	0.71

Correlation Heatmap